

Proof by Mathematical Induction



The three steps of induction

- 1 State the three steps required for a valid proof by mathematical induction.
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Proving a summation formula

- 2 Prove by induction that $\sum_{r=1}^n r = \frac{n(n+1)}{2}$ for all $n \geq 1$.
 - 3 Prove by induction that $\sum_{r=1}^n r^2 = \frac{n(n+1)(2n+1)}{6}$ for all $n \geq 1$.
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Proving divisibility results

- 4 Prove by induction that $7^n - 1$ is divisible by 6 for all $n \geq 1$.
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Induction with inequalities

- 5 Prove by induction that $2^n > n^2$ for all integers $n \geq 5$.
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Induction in calculus: derivatives of a product

- 6 Prove by induction that $\frac{d^n}{dx^n}(e^{ax}) = a^n e^{ax}$ for all positive integers n .
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Induction with a recursively defined sequence

- 7 A sequence is defined by $u_1 = 3$ and $u_{n+1} = 2u_n - 1$. Prove by induction that $u_n = 2^n + 1$ for all $n \geq 1$.
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Proving a geometric-series-style summation

- 8 Prove by induction that the sum $3^0 + 3^1 + \dots + 3^{n-1}$ equals $\frac{3^n - 1}{2}$ for all $n \geq 1$.
 - 9 Prove by induction that the sum $1 \cdot 1! + 2 \cdot 2! + \dots + n \cdot n!$ equals $(n+1)! - 1$ for all $n \geq 1$.
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Proving an inequality involving factorials

- 10 Prove by induction that $n! > 2^n$ for all integers $n \geq 4$.
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Proving a divisibility result with two variables

- 11 Prove by induction that $n^3 + 2n$ is divisible by 3 for all positive integers $n \geq 1$.
- 12 Prove by induction that $9^n - 1$ is divisible by 8 for all positive integers $n \geq 1$, and explain briefly why a similar argument would show $a^n - 1$ is divisible by $a - 1$ for any integer $a > 1$.
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Induction with a recurrence relation for a sequence of sums

- 13 A sequence satisfies $S_1 = 2$ and $S_{n+1} = S_n + 6n + 2$ for $n \geq 1$. Prove by induction that $S_n = 3n^2 - n$ for all $n \geq 1$.
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Induction with a strengthened hypothesis

- 14 Prove by induction that $11^n - 4^n$ is divisible by 7 for all positive integers $n \geq 1$.
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Induction to prove a formula for the sum of cubes

- 15 Prove by induction that $\sum_{r=1}^n r^3 = \left[\frac{n(n+1)}{2} \right]^2$ for all $n \geq 1$.
- 16 Prove by induction that $5^n + 2 \cdot 3^{n-1} + 1$ is divisible by 8 for all positive integers $n \geq 1$.
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Induction to prove a formula for a weighted geometric sum

- 17 Prove by induction that $\sum_{r=1}^n r \cdot 2^r = (n-1)2^{n+1} + 2$ for all $n \geq 1$.
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A closing check: identifying an invalid induction argument

- 18** A student claims to prove that all horses are the same color using induction on the number of horses n in a group: the base case $n = 1$ is trivially true (one horse is trivially "the same color" as itself), and the student then argues that if any k horses are the same color, then any $k + 1$ horses must also be the same color. Explain precisely where this argument breaks down, and state why a correct induction proof must avoid this flaw.